

SAGAR CHAUDHARY

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SUMMARY

Backend Engineer building distributed, event-driven systems in Python and Go. Architected real-time data pipelines ingesting high-frequency sensor streams over raw TCP/IP sockets and delivering them to clients over WebSockets at sub-second latency, with asynchronous processing and multithreaded batched writes that sustain continuous high-volume load. Designs microservice and publish-subscribe architectures, models normalized PostgreSQL schemas alongside NoSQL stores, and secures APIs with JWT and role-based access control. Ships with Docker, Kubernetes, CI/CD and full observability - plus applied ML and GenAI experience for teams building AI-backed products.

TECHNICAL SKILLS

Backend & Data Pipelines: Python, Go, Java, FastAPI, REST APIs, WebSockets, TCP/IP sockets, microservices, event-driven pub-sub, Redis Streams, asyncio, JWT, RBAC

Databases & Cloud: PostgreSQL (SQL), NoSQL (MongoDB), MySQL, ChromaDB, AWS, Linux, Git

MLOps & Deployment: Docker, Kubernetes, KEDA autoscaling, CI/CD (GitHub Actions), Prometheus, OpenTelemetry, ONNX Runtime model serving, structured logging, monitoring, crash-recovery design

Machine Learning: classification, regression, time-series forecasting (LSTM, ARIMA), ensembles (XGBoost, LightGBM, AdaBoost), feature engineering, data cleaning & pre-processing, SMOTE, Optuna, Scikit-Learn, Pandas, NumPy, Statsmodels

Deep Learning: PyTorch, TensorFlow/Keras, CNNs (YOLO, Detectron2, ZoeDepth), LSTM/RNN time-series models, BERT/Transformers, transfer learning, LiDAR-camera sensor fusion, OpenCV, Open3D

Generative AI & Agentic: LangChain, LangGraph, RAG, agentic AI (tool-use agents, MCP), chunking strategies, prompt engineering, prompt-to-template design, rerankers, vector search (ChromaDB), LLM & embedding APIs (OpenAI, Gemini, Ollama, Nomic)

Core CS: Data Structures, OOP, Distributed Systems, System Design, Concurrency, Operating Systems, Computer Networks

EXPERIENCE

Deevia Software, Bengaluru

Apr 2025 - Present

Graduate Trainee Engineer

Nov 2025 - Present

Real-Time Engine Test-Bed Monitoring & Telemetry Platform (Industrial IT-OT System)

Tech Stack: Python, FastAPI, LangChain, LangGraph, LSTM (PyTorch), RAG, ChromaDB, Ollama, OpenAI, Gemini, WebSockets, TCP/IP Sockets, PostgreSQL, React.js, TypeScript, Redux Toolkit, Docker, JWT, RBAC

- Architected an end-to-end IT-OT data pipeline for an engine test facility - ingesting high-frequency industrial sensor data over raw TCP/IP sockets and streaming it to live dashboards via WebSockets at sub-second latency.
- Trained LSTM deep-learning models on engine sensor time-series for trend forecasting, anomaly detection, and predictive-maintenance insights during live test runs.
- Optimized ingestion with asynchronous processing and multithreaded batched database writes, sustaining continuous high-volume sensor load without memory backpressure.
- Designed an event-driven publish-subscribe architecture that decoupled ingestion, processing, and streaming services, so new sensor types are added through configuration rather than code changes.
- Modeled normalized PostgreSQL schemas and tuned SQL queries for fast retrieval of historical test runs, ensuring data integrity across ingestion and storage layers.
- Implemented automated crash-recovery and graceful-shutdown handling for clean restarts without data loss.

Multi-Vendor Camera Device Management Framework

Tech Stack: Python, FastAPI, PostgreSQL, MongoDB (NoSQL), React, Docker, CI/CD, OOP Design Patterns

- Built a device management framework using object-oriented design - abstract base classes and interface layers - to abstract camera hardware, enabling drop-in integration of multiple camera vendors without backend changes.
- Developed a FastAPI backend exposing REST endpoints for device registration, configuration management, and real-time status monitoring across distributed devices.
- Designed a dual-database layer - PostgreSQL for relational device records and MongoDB (NoSQL) for heterogeneous, schema-less device configurations - to handle structured and unstructured data efficiently.

Machine Learning Intern

Apr 2025 - Nov 2025

Distributed Multi-Sensor Video Analytics Platform

Tech Stack: Go, Python, FastAPI, REST APIs, WebSockets, AWS, PostgreSQL, Docker, Microservices, OpenCV

- Designed a distributed microservices architecture on AWS separating ingestion, processing, and streaming layers for scalable multi-camera plus sensor pipelines.
- Built a dual-backend system: Go services handling real-time API orchestration and request validation, with Python services running compute-intensive video processing.

- Built a multi-format ingestion pipeline supporting heterogeneous camera systems with mixed inputs (video, CSV, binary sensor files).
- Used WebSocket-based inter-service messaging for low-latency async job status and result streaming; containerized all services with Docker for reproducible deployment across environments.

Foot Monitoring System (ADAS)

Tech Stack: Python, PyTorch, YOLOv11, Detectron2, ZoeDepth, OpenCV, FastAPI, React, PostgreSQL, AWS

- Built a FastAPI backend with REST endpoints for data access, annotation workflows, and real-time inference, with abstract interfaces for multi-camera modularity.
- Automated data collection, labeling, and continuous model retraining using Dockerized microservices and open-source annotation tools (LabelMe, LabelImg).

3D Object Detection via LiDAR-Camera Sensor Fusion (KITTI)

Tech Stack: Python, PyTorch, YOLOv8 (CNN), SFA3D, OpenCV, Open3D, NumPy, SLAM, RANSAC/PROSAC, Docker

- Built a multi-modal 3D object detection system fusing LiDAR point-cloud detections (SFA3D) with CNN-based 2D camera detections (YOLOv8) for robust perception in autonomous-driving scenes.
- Designed a confidence-weighted fusion algorithm that combines per-sensor detections by IoU association, improving accuracy and robustness over single-sensor baselines - 92.44% / 80.23% AP (easy / hard) on the KITTI benchmark.

PROJECTS

Inferno - Distributed ML Inference & Agentic AI Platform ([Live Demo](#) · [GitHub](#))

Tech Stack: Python, FastAPI, Redis (Streams + Pub/Sub), WebSockets, asyncio, PyTorch, ONNX Runtime, YOLOv8, faster-whisper, Sentence-Transformers (RAG), MCP (Agentic AI), Prometheus, OpenTelemetry, React, TypeScript, Docker, Kubernetes, KEDA, GitHub Actions

- Built a horizontally-scalable model-serving platform - a stateless async FastAPI gateway, a Redis-Streams broker, and a pool of shared-nothing worker processes - that batches inference requests and streams results back over WebSockets to a live ops dashboard, solving the core serving problem of high throughput and low-latency real-time delivery.
- Implemented dynamic request batching (size-or-timeout window) that amortizes per-call model overhead and keeps the GPU/CPU saturated - raising throughput under load while capping tail latency at the batch-wait window, with no manual tuning.
- Engineered backpressure with hysteresis (HTTP 429 + Retry-After, dual high/low water-marks) so the platform sheds load and stays stable under overload instead of collapsing into latency spikes and OOM.
- Guaranteed zero job loss using Redis Streams consumer groups with a publish-before-ack ordering - dead-worker jobs are reclaimed via XPENDING/XCLAIM (at-least-once delivery with idempotent, TTL-bounded result keys).
- Scaled WebSocket result delivery to O(1) Redis connections via a single-connection PSUBSCRIBE result router, eliminating the connection-pool exhaustion that capped the naive design at ~120 concurrent clients.

Customer Churn Prediction

Tech Stack: Python, Pandas, Scikit-Learn, XGBoost, LightGBM, AdaBoost, Optuna, SMOTE

- Handled class imbalance with SMOTE and tuned hyperparameters with Optuna - 94.6% accuracy, 0.96 AUC-ROC, 93.2% precision.

ACHIEVEMENTS & CERTIFICATIONS

- Top 10 Finalist - HumanAlze YouData.ai Hackathon (out of 150 teams); built a scalable NLP pipeline for document classification. Hack2skill certificate (ID: 2024H2S06EH10008).
- 5th Place - CODE RED 2024 Hackathon (24-hour event); led a team of 4 to build a marine-life classification system using CNNs, real-time image processing, and transfer learning.
- Machine Learning Specialization - Stanford / DeepLearning.AI (Coursera), Dec 2023
- Neural Networks and Deep Learning - DeepLearning.AI, May 2024 · OpenCV Bootcamp - OpenCV University, Apr 2025 (Grade: 100%)

EDUCATION

Bangalore Institute of Technology — B.E. in Information Science and Engineering (CGPA: 8.6)

Dec 2021 - Jul 2025

DAV Public School, Jamshedpur — CBSE Class 12: 92.4%

Apr 2020 - May 2021

DAV Public School, Jamshedpur — CBSE Class 10: 92.4%

Apr 2018 - Apr 2019